

Iraklis Sarigiannidis

Born on November 24, 1997, in Katerini, Greece.

Currently pursuing a PhD in Paris, France.

✉ Mail 🔗 LinkedIn 🌐 Website

High-frequency electromagnetic simulation engineer with strong expertise in wave propagation, scattering, and radar signal modeling. Experienced in numerical methods (MoM, FEM, FDM) and proficient in Matlab, Python, and Comsol/Feko for acoustic and electromagnetic wave modeling.

Key Skills

- Electromagnetic field modeling and wave scattering
- High-frequency techniques and radar signal simulation
- Computational methods : MoM, FEM, FDM
- Signal processing, data analysis, and algorithm development
- Software : Matlab, Python, Comsol, Altair Feko, LaTeX

Education



- **International joint PhD supervision** (Dec 2022 – Dec 2025)

High-frequency techniques for electromagnetic diffraction and antenna problems.

ED SMAER 391, L'école doctorale Sciences Mécaniques, Acoustique, Électronique, et Robotique de Paris.

- ▷ GeePs laboratory - Science Sorbonne University - Paris - France (Dec 2022 – Sept 2024)
- ▷ Laboratory of applied electromagnetics - University of Siena - Siena - Italy (Sept 2024 – Dec 2025)

- **Master 2 WaveWiCom** (Sept. 2022 – Dec 2022)

Waves and Devices for Advanced Wireless Communication Systems

- ▷ Science Sorbonne University

- **Master Thesis** (Sept. 2020 – Dec. 2021)

(Grade 10/10 Excellent, ECTS: 30 units)

Bi-sphere monostatic electromagnetic scattering cross section.

- ▷ Aristotle University of Thessaloniki

- **Integrated Master's Degree** (Dec. 2018 – Sept. 2020)

Specialization area **Telecommunications**

- ▷ Aristotle University of Thessaloniki

- **Electrical and Computer Engineering** (Sept. 2015 – Dec. 2018)

(Overall grade: 8.99/10 Excellent, Total ECTS: 329 units)

- ▷ Aristotle University of Thessaloniki

Employment



- **Researcher Assistant** (Feb. 2022 – Sept. 2022)

Remote sensing for characterizing forest and rural road network.

- ▷ Visual Computing Lab, CERTH/ITI, collaborations: kartECO, AUTH

- **Bioengineering Intern** (Mar. 2020 – Jun. 2020)

Advanced processing for spike sorting of neural activity.

▷ NextGrowth Novelty Corporation, collaboration : Medical School AUTH.

Scholarships

- Awarded the **Franco-Hellenic Cooperation Scholarship 2022–2023** for Master 2 studies in France. (Sept. 2022 – Dec. 2022)
- Awarded **PhD public funding** in France through the **ED SMAER 391 competition**. (Dec. 2022 – Dec. 2025)

Research Publications

Journal Articles

- **COMPEL, Emerald group publishing** *Accepted for publication (in press)*
Integral Equation Numerical Approach for the Determination of Diffraction Coefficients from Generally Shaped Cones.

Conference Presentations

- **ICEEA 2023 Venice, Italy** (09 – 13 Oct. 2023)
Conference presentation and abstract paper publication
I. Sarigiannidis, M. Casaletti, and M. Albani, (2023). A hybrid analytical-numerical approach for the calculation of the GTD diffraction coefficient for the diffraction at the tip of an arbitrary section cone. Presented at the International Conference on Electromagnetics in Advanced Applications (ICEAA), Venice, Italy. [DOI : 10.1109/ICEAA57318.2023.10297755]
- **Numelec Toulouse, France** (08 – 10 Jul. 2024)
Conference poster
Integral Equation Numerical Approach for the Determination of Diffraction Coefficients from Generally Shaped Cones.
- **EMTS 2025 Bologna, Italy** (23 – 27 Jun. 2025)
Conference presentation
Semi-Analytical Formulation of Diffraction from a Cone with an Arbitrary Cross-Section Using FEM.
- **IEEE AP-S/URSI 2025 Ottawa, Canada** (13 – 18 Jul. 2025)
Conference presentation and paper publication
Semi-Analytical Computation of the Scattering Field for Arbitrarily Shaped Semi-Infinite Cones.
- **ICEAA 2025 Palermo, Italy** (8 – 12 Sept. 2025) *Accepted*
Conference presentation
Semi-Analytical Formulation of Diffraction from a Cone with an Arbitrary Cross-Section.

Computational Methods

Custom implementations of • Method of Moments (MoM) • Finite Element Methods (FEM) • Finite Difference Methods (FDM) • Physical Optics (PO) • Stationary Phase Method (SPM)

Languages

- Greek (native speaker)
- English (B2 Edexcel University)
- French (B1 Sorbonne University)
- Italian (A2 University of Siena)

Other Skills

- Academic training, research and teaching
- L^AT_EX typesetting and publication

Miscellaneous Experience

- **Teaching Assistant**

Laboratory sessions for electromagnetic simulation and antenna design using Altair Feko (17 total hours).

- **European Microwave Week Berlin, Germany** (17 – 22 Sept. 2023)

Volunteering and attendance

- **European School of Antennas**

▷ Sorbonne University (25 – 28 Sept. 2023)

Exploiting symmetries in artificial materials for antenna applications.

▷ University of Siena (30 Sept. – 4 Oct. 2023)

Reconfigurable intelligent surfaces for smart radio environment.

- **Speaking in public seminar**

▷ Sorbonne University (15 – 16 Apr. 2024)